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Appendix

(Note: some completions contain unpleasant content, including slurs.)

A Broader Impacts

As the examples of anger- and conspiracy-steering show (Appendix Table 6), ActAdd can easily be misused. Insofar as existing methods for steering LLMs leave the target goal or property somewhere 'in' the model (but simply make sampling it low probability) Lyu et al. 2024, activation engineering may circumvent superficial alignment methods.

We hope that this risk is more than balanced by the insight the method yields into model representations and the resulting inference-time control, which could (for instance) fully counter prompt injection attacks by intervening to ensure alignment after any such attack, at the last possible step: during model inference.

B Is ActAdd just a subtle kind of prompt engineering?

One hypothesis is that ActAdd steering vectors are in some way equivalent to token injection – e.g. adding a virtual ' weddings' token at the given stream position. This is plausible for simpler interventions. Given the prompt 'I love you because', if we inject a ' wedding' token into the first residual stream with a large coefficient, perhaps the model indeed just processes the prompt as ' wedding love you because' instead.

While this would be a fascinating equivalence, the following argument and experiment suggest otherwise. Since tokens are discrete, the token injection hypothesis comes apart from the linear representations hypothesis in cases like adding $3 \times$ '$\mathrm{wedding}$' and then $-3 \times$ '$<\mathrm{whitespace}>$', on top of the token 'I'. Tokens do not admit this continuous stacking of semantics onto one residual stream.

However, consider the steering vector for `Anger− Calm` with $l = 20, c = +10$. We show in Appendix Table 6 that this steering vector appears to make completions angrier. Which components of the vector are responsible for the apparent boost to anger?

Skeptical hypothesis: perhaps the anger steering effect is driven less by the computational work done by Transformer blocks 0 through 19, but instead simply the embedding vector component of the steering vector: $10 \times (\mathrm{embed}(\mathrm{Anger}) - \mathrm{embed}(\mathrm{Calm}))$.

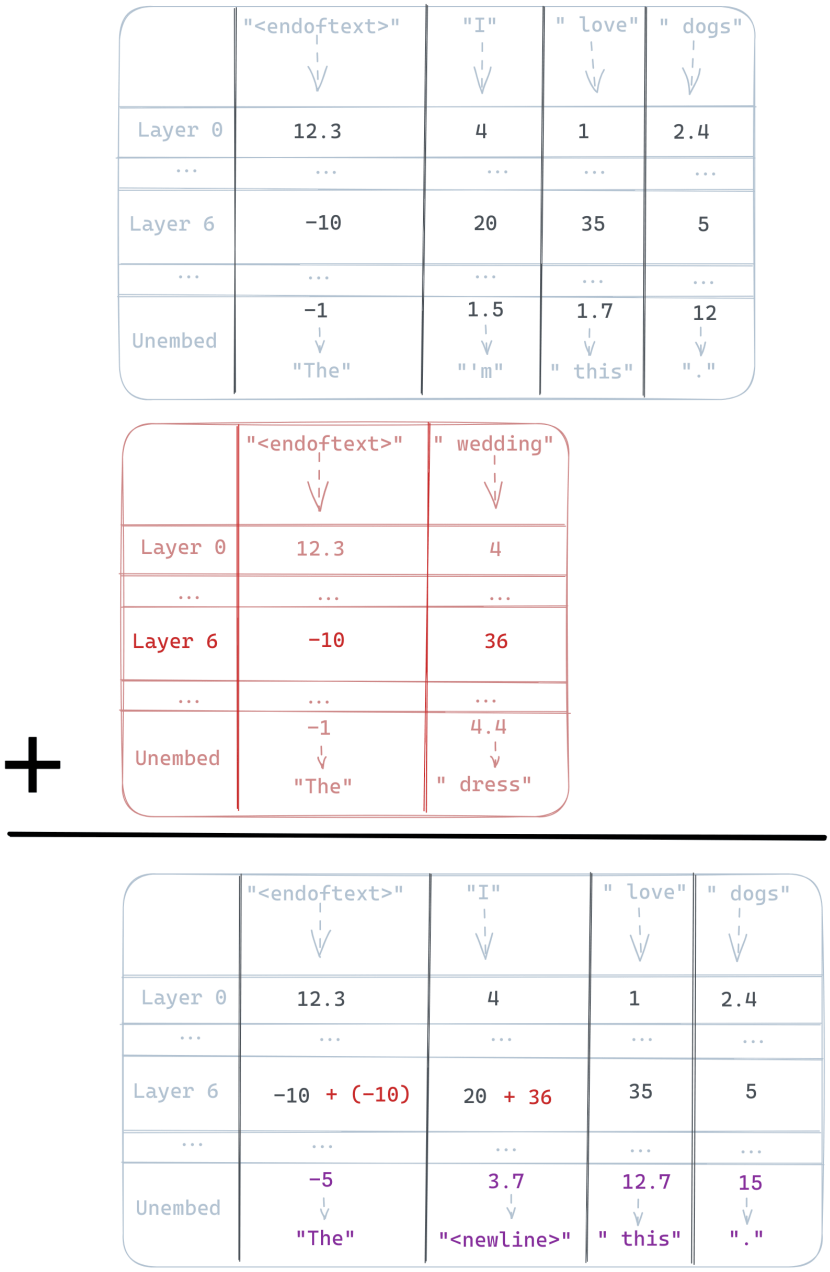

Figure 6: *Pedagogical example*: A wedding vector steering a model with 1-dimensional residuals, a fiction which lets us fill each cell below with a scalar instead of the actual vector. Let the user prompt $p^* =$ 'I love dogs'. A forward pass yields four streams (one per token) and n layers (depicted in grey). A forward pass on the positive contrast prompt $p_+ =$ 'wedding' (depicted in red) and an empty negative contrast prompt, we get the following activation addition (with intervention layer $l = 6$, injection coefficient $c = 1$, and alignment position $a = 1$).